TICAGRELOR

Dyspnoea common

Twice daily

CHEMICAL NAME (1S,2S,3R,5S)-3-[7-[[(1R,2S)-2-(3,4-difluorophenyl)cyclopropyl]amino]-5-(propylthio)-3H-[1,2,3]triazolo[4,5-d]pyrimidin-3-yl]-5-(2-hydroxyethoxy)cyclopentane-1,2-diol

CLASS Reversible P2Y12 receptor antagonist (cyclopentyltriazolopyrimidine)

TRADE NAMES Brilinta · Brilique · Possia

THERAPEUTIC USES / INDICATIONS

- Acute coronary syndrome (STEMI, NSTEMI, unstable angina) with aspirin; superior to clopidogrel (PLATO).
- Long-term secondary prevention after MI (60 mg dose).
- Minor stroke/high-risk TIA (with aspirin, short-term).

ADMINISTRATION

- **PO** twice daily with or without food; tablets can be crushed and given via NG tube.
- Take with low-dose aspirin (aspirin ≤100 mg/day).

DOSAGES

- **Loading:** 180 mg single dose.
- **Maintenance:** 90 mg twice daily for 12 months.
- **After 12 months (high risk):** 60 mg twice daily.
- Aspirin 75–100 mg daily alongside.

PHARMACOKINETICS

- **Onset:** 30 min; peak 2 h. **Duration:** 3–5 days (reversible binding).
- Active itself (not a prodrug); metabolised by CYP3A4 to active metabolite.
- Excreted in bile. **t½:** 7–9 h.

NURSING MANAGEMENT

- Monitor for bleeding and CBC.
- Assess breathlessness; distinguish from heart failure; report.
- Check HR and rhythm.
- Ensure aspirin dose ≤100 mg.
- Reinforce twice-daily adherence.

PHARMACODYNAMICS

- Reversibly blocks **P2Y12** receptor → faster, stronger platelet inhibition than clopidogrel.
- Inhibits adenosine reuptake (↑ adenosine → dyspnoea, bradyarrhythmias).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dyspnoea (shortness of breath, usually mild and transient).
- Bruising, minor bleeding.
- Headache, dizziness.

ADVERSE EFFECTS (SERIOUS)

- Major and fatal bleeding.
- Ventricular pauses, bradyarrhythmias.
- Raised uric acid, creatinine.
- Central sleep apnoea (rare).

CONTRAINDICATIONS & CAUTIONS

- Active pathological bleeding; history of intracranial haemorrhage.
- Severe hepatic impairment.
- Use with strong CYP3A4 inhibitors (ketoconazole, clarithromycin, ritonavir).

KEY INTERACTIONS

- **Aspirin >100 mg/day:** ↓ efficacy.
- **Strong CYP3A4 inhibitors:** contraindicated; **inducers** (rifampicin): ↓ effect.
- **Simvastatin/lovastatin >40 mg:** ↑ levels.
- **Digoxin:** ↑ levels; **anticoagulants:** ↑ bleeding.

PATIENT EDUCATION

- Take twice daily — missing doses increases heart attack risk.
- Mild breathlessness is common and often settles; report if severe.
- Do not stop without your cardiologist.
- Report bleeding or black stools.

KEY POINTS

- Reversible P2Y12 inhibitor, more potent than clopidogrel.
- Dyspnoea side effect.
- 180 mg load, 90 mg BD.
- Aspirin must be ≤100 mg/day.

SOURCES CHECKED [409] Ticagrelor 60 mg film coated tablets (UK SmPC) [410] Ticagrelor, oral (US label) [411] Ticagrelor, oral (US label, Alembic Pharmaceuticals Limited)

Full references p. 432–457